

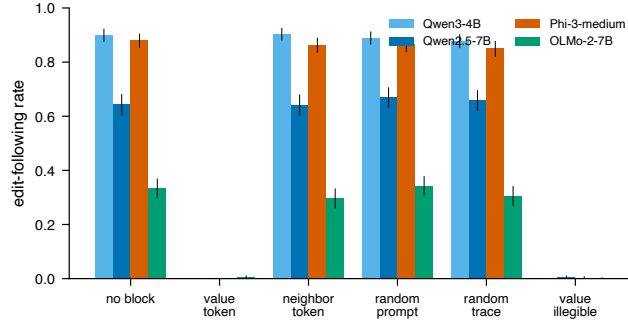


Figure 2: Rate at which the answer follows the edited value under each blocked attention target. Blocking the value’s own position collapses following on every model; matched control blocks do not. Error bars are 95 percent bootstrap intervals.

Table 2: Attention block on arithmetic chains: rate at which the answer follows the edited value. OLMo-2-7B continues this format less reliably (low no-block rate) and shows the same contrast.

model	no block	value position	neighbor	random position	earlier value
Qwen3-4B	0.900	0.000	0.903	0.778	—
Qwen2.5-7B-Instruct	0.643	0.000	0.640	0.670	0.658
Phi-3-medium	0.880	0.000	0.863	0.865	0.850
OLMo-2-7B	0.333	0.005	0.297	0.342	0.305

words, of a random prompt position, or of an earlier step’s written value leave the dependence at its unblocked level. Blocking the operand the next step needs breaks the arithmetic itself, yielding answers that match neither the edit nor the correct value. The collapse holds across three families and on both task formats (natural-language task: 0.730 to 0.007 on Qwen3-4B, 0.720 to 0.003 on Phi-3-medium).

The block removes the value, not the ability to generate. Under the value-position block the unparseable-output rate stays at 0.000 (0.040 on OLMo-2-7B, equal to its unblocked rate). The model produces a fluent, parseable number that matches neither the edited value nor the correct one: it computes forward from a value it can no longer retrieve.

Retrieval occurs in the middle third of the network. Layer-resolved versions of both interventions locate the read in depth on Qwen3-4B (36 layers). Overwriting the value’s hidden state at any single layer from 0 through 22 redirects the answer at 0.90 to 0.94, matching the full multi-layer patch (0.92), while single-layer patches at layer 24 and beyond redirect at 0.02 or less ($n = 60$): the register can be rewritten anywhere upstream of its consumers and nowhere downstream of them. Blocking the value-position attention only at the middle third of layers (12 through 23) collapses following from 0.857 to 0.025, blocking only the early or late third leaves it unchanged (0.857 and 0.863), and no single layer’s block matters (0.84 to 0.89 at every layer, $n = 80$): the read is distributed redundantly across the middle third’s layers and confined to it. The two sweeps agree on the same boundary, near layer 23: the value is written into the position’s residual stream from early depth, retrieved by attention across the middle third, and causally inert afterward. Coarse sweeps on two further models (appendix) show the structure generalizes while the boundary’s depth is model-specific: on Phi-3-medium, early- and mid-third patches redirect at 0.89 and 0.87 and late-third patches at 0.007, matching the Qwen3-4B shape, while on Qwen2.5-7B patches in every third redirect at 0.70 to 0.73, so its reads extend into the final third of its 28 layers.

Reading is the preferred path, not the only one. Replacing the value’s characters with unreadable dots, without touching attention, also removes following, and differently by model: the Qwen models reconstruct the value from the previous line and recover the correct answer in 0.60 to 0.73 of cases, while Phi-3-medium does not recover (0.00).

5 What a calibrated J-space ablation does and does not disturb

Two results relate the written registers to the model’s internal capacity. First, the internal side cannot replace writing on these tasks: instructed to output only the final answer, with token counts confirming nothing else was generated, models fail beyond one to two dependent steps (Qwen2.5-7B falls from 1.00 at $d = 1$ to 0.04 at $d = 4$; DeepSeek V3.2 at 671B reaches 0.23 at $d = 1$ and near zero above), and models write from the first step rather than upon overflow (0.93 to 1.00 of ground-truth intermediates appear in free traces at every difficulty, for every model from 1.5B to 671B).

Second, an ablation of J-space, the concept subspace a fitted Jacobian lens reads from the residual stream [Gurnee et al., 2026], leaves the externalized computation intact. At the strongest dose that preserves ordinary generation (calibrated and frozen on neutral text: next-token agreement 0.87, perplexity ratio 1.12), removing the top 16 active concepts at every position changes neither chain-of-thought accuracy nor bare-answer accuracy nor the amount written, on synthetic chains and on GSM8K, and does no more than a size-matched random-directions ablation. We read this as a boundary on the particular readout tested, for three reasons. The dose is capped at generation-preserving strength, the lens is one fitted artifact, and a written token’s contextual representation and its J-space component are not exclusive alternatives, since the lens samples the same residual states our patch overwrites. The J-space account itself reports that explicit chain of thought makes reasoning more robust to J-space ablation and interprets writing as externalizing state the model would otherwise carry internally [Gurnee et al., 2026]. Our result is consistent with that reading and sharpens it with a mechanism: on these tasks the externalized channel is where the serial state causally resides, and the tested lens readout does not capture it.

6 When the written value is read versus recomputed

Models read the register by default. On DeepSeek V3.2, an edited value the prompt merely implies is followed at 0.93, while an edited value the prompt generatively defines (the prompt states the two numbers whose sum becomes the running value) is reverted to correct at 0.43; with the defining sentence present but the edit two steps away, following returns to 0.96, so the reversion targets exactly the defined value. Claude Sonnet 4.5 recomputes more broadly, reverting most when a stated checkpoint is the value’s only source. Llama 3.3 70B follows edits at 0.97 whether or not the value is prompt-defined, despite solving the same chains from scratch, so the capacity to recompute does not imply recomputing. Qwen models at 4B and 7B never revert. These conditions come from separate experiments with different trace lengths and item sets; each within-experiment contrast is controlled, and we do not claim a single ordered scale across models or that reconstruction cost is the governing rule. Characterizing what governs the choice is open.

The same test bounds the mechanism on natural benchmarks: on GSM8K worked solutions, an edited intermediate is followed at 0.87 by Qwen3-4B and at 0.10 by V3.2, since a benchmark intermediate is often one operation from numbers still in the prompt. Written values act as memory chiefly for state the model does not reconstruct on its own.

7 Related work

Our interventions build on causal tracing and activation patching [Meng et al., 2022] and causal abstraction [Geiger et al., 2021]; Heimersheim and Nanda [2024] caution that broad patches conflate a component mattering with deleting information at a position, which the single-layer sweep in Section 4 addresses. That written intermediates can serve as memory is established: theoretically [Merrill and Sabharwal, 2024], behaviorally [Lanchantin et al., 2023], and causally in specific settings [Zhu et al., 2025, Shih et al., 2026]. We add that the behavior is native to unmodified pre-trained models across four families and two formats, the attention-level retrieval route with matched controls, its depth localization, register composition, the qualified J-space boundary, and the read-versus-recompute map. Faithfulness studies show stated reasoning and internal computation can diverge [Turpin et al., 2023, Lanham et al., 2023]; filler-token results show computation can pass through uninformative tokens [Pfau et al., 2024]; both bound our claims to state the model does not regenerate internally.

8 Limitations

Tasks are numeric and count state tracking, so our claims concern serial task state rather than working memory in general; code, plans, and multi-entity state are untested. The J-space result bounds one fitted lens at generation-preserving doses and does not measure the J-space content of the written positions themselves. The fine layer sweeps are established on one model; the cross-model sweeps are coarse, and on Qwen2.5-7B they do not resolve the read boundary. White-box evidence is at 4B to 14B; frontier evidence is behavioral. The read-versus-recompute results support within-experiment contrasts, and the only patch control is size-matched noise; a structured control planting a valid activation for a different quantity would further isolate value content.

9 Conclusion

On serial state-tracking tasks, unmodified pretrained language models spontaneously use the hidden states at written intermediate-value positions as compositional registers: settable by intervention, retrieved through attention before two-thirds depth, independently addressable, and decisive for the answer. A calibrated J-space ablation leaves this externalized computation undisturbed, which bounds that readout on these tasks. For oversight the implication cuts both ways. The written positions are a causally used memory channel, so the trace exposes the symbols the computation operates on; and identical visible text can coexist with different stored values and different downstream behavior, so the text does not guarantee the hidden content at those positions. Trace monitoring observes the channel, and verifying the state behind it is a separate problem.

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